

# Where is technician capacity being **wasted**—and what should dispatch do about it this week?

A star-schema SQL model, a formula-driven Excel operational control, Power BI measures, and a live dashboard, built on two years of synthetic HVAC/electrical dispatch data for Ridgeline Field Services.

PREPARED FOR

**Dispatch & Operations  
Leadership**

REPORTING

WINDOW

**Jan 2024 –  
Dec 2025**

DATASET

**37,404 job  
records**

STACK

**SQL Server · Power  
BI · Excel**

**Data disclosure.** Ridgeline Field Services is a fictional company built for this portfolio project. Every figure in this document is derived from synthetic data generated by `sql/02_generate_synthetic_data.sql`. No confidential, proprietary, or real-organization data is used or represented anywhere in this analysis.

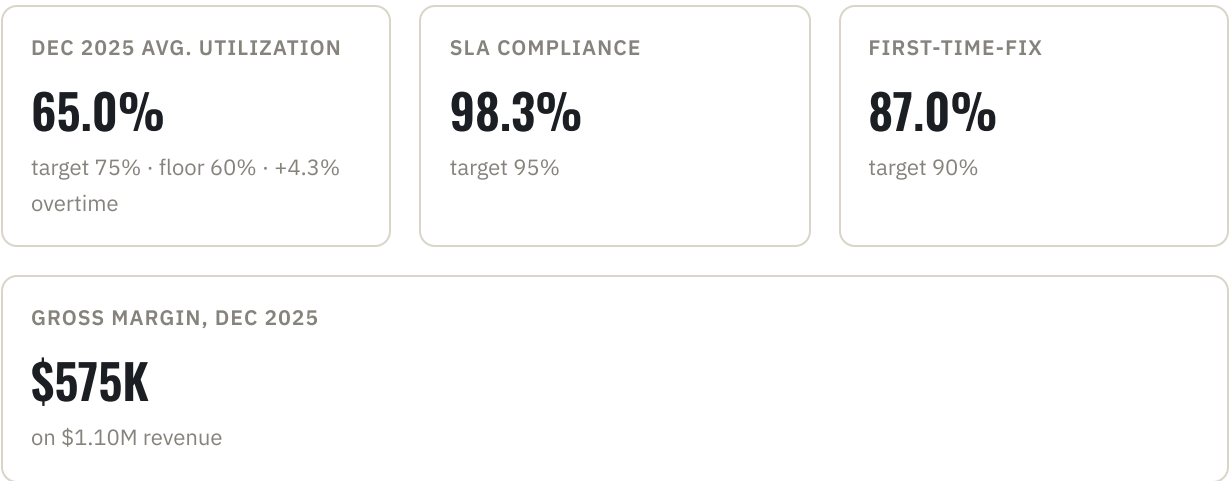
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# 01 · Executive summary

Ridgeline Field Services dispatches HVAC and electrical technicians across five regions. Leadership's standing question was simple to ask and, until this project, expensive to answer with a spreadsheet: **where is technician capacity sitting idle, and where is it overcommitted — and what, specifically, should change?**



Three findings drive the recommendations in this report:

- 1. **South Metro's utilization never once overlaps with any other region's** across the full two-year window (41.1%–55.3%, vs. 59.3%+ everywhere else) — a structural staffing/dispatch imbalance, not a one-off dip.
- 2. **East Valley's SLA problem is invisible in the aggregate number** (97.9% overall) and only appears when urgent, 1-hour-window jobs are isolated (93.2%) — a travel-distance-driven gap that a single blended KPI would hide.
- 3. **Apprentice technicians run a callback rate roughly 4.7x higher than Master technicians** (21.6% vs 4.6%), concentrated enough to show up repeatedly in the priority action queue below.

All three are backed by a reusable SQL layer, cross-validated against an independently-built Excel formula layer — a cross-check that caught two real bugs during this project's build (Section 08), which is the actual argument for building the same logic twice rather than once.

## 02 · Business question & scope

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**Business question:** Where is technician capacity being wasted, and which dispatch decisions should change this week?

Scope: five regions (North Metro, South Metro, East Valley, West Hills, Central District), ten HVAC/Electrical service types, 40 technicians (35 active, 5 inactive in this generation run), 300 clients, and 37,404 dispatched jobs across a 24-month window (Jan 2024–Dec 2025). Out of scope: real-time dispatch routing/GPS optimization, and multi-year demand forecasting — this is a monthly-refresh analytical model built to support a weekly dispatch-review cadence, not a live routing engine.

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## 03 · Methodology & data validation

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The analytical layer is a five-dimension/one-fact star schema in SQL Server ( `Dim_Date` , `Dim_Region` , `Dim_Technician` , `Dim_Client` , `Dim_ServiceType` , `Fact_ServiceJobs` ), plus a `Ref_KPITargets` table that centralizes every threshold used downstream — change a number there and the SQL views, the Excel dashboard, and the Power BI conditional formatting all move together.

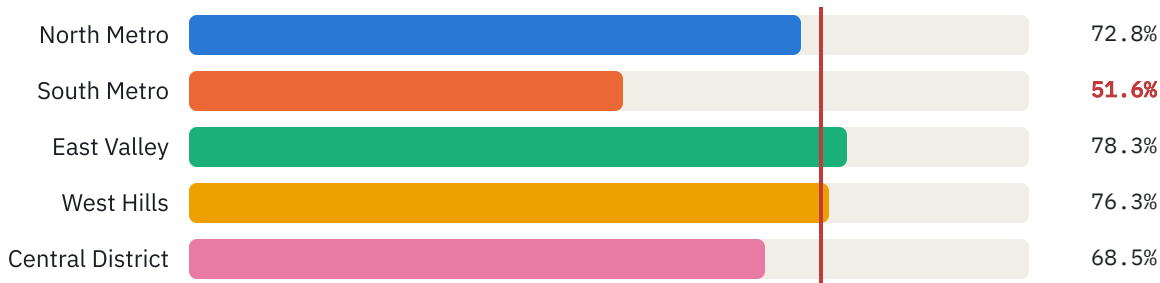
The dataset is **synthetic by design, not by necessity** — a set-based T-SQL generator built in deliberate, documented imbalances (region workload intensity, travel-time baselines, skill-driven fix probabilities, HVAC/Electrical seasonality) so the KPI layer would have something real to find. It was then profiled — row counts, referential integrity, distribution sanity checks — before any dashboard was built on top of it. The full write-up, including a generation bug the profiling step caught (technicians dispatched before their hire date), is in `docs/data_validation_report.md`.

Every KPI is implemented as a parameterized SQL view/procedure ( `sql/04_kpi_views.sql` , `sql/05_stored_procedures.sql` ) and then **re-implemented independently** as live formulas in the Excel workbook — SUMIFS/AVERAGEIFS against a 37,404-row table, not a paste of query results. The two are cross-checked against each other rather than assumed to agree (Section 08).

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## 04 · Finding 1 — Regional capacity imbalance

Utilization is measured as (job duration + travel time) actually worked, divided by available weekday capacity, per technician per month. December 2025:



Red marker = 75% target. Source: `dbo.vw_RegionKPIMonthly`, cross-checked against `KPI_Dashboard` in the Excel workbook (exact match).

**This is not a one-month anomaly.** Across all 24 months, South Metro's utilization ranges 41.1%–55.3% and **never once overlaps** with any other region's range (the next-lowest region, West Hills, never drops below 59.3%). South Metro is authorized for 8 technicians against a workload that consistently supports fewer. At the other end, East Valley averages the highest utilization of the five regions (79.2%), and several of its individual technicians post >95% utilization in a given month — an over-capacity/burnout risk (see the priority queue in Section 07).

## 05 · Finding 2 — Travel-driven SLA risk (a segment, not an aggregate)

SLA here is an **arrival-window commitment** (e.g. "technician arrives within 1 hour of the scheduled window"), not a multi-day completion deadline — this is how most residential/commercial field-service SLAs are actually written, and it is what makes dispatch delay and travel time SLA-relevant in the first place.

| REGION           | OVERALL SLA | 1-HR URGENT-RESPONSE SLA | GAP     |
|------------------|-------------|--------------------------|---------|
| Central District | 98.6%       | 95.3%                    | 3.3 pts |
| South Metro      | 98.5%       | 95.0%                    | 3.5 pts |
| North Metro      | 98.3%       | 94.6%                    | 3.7 pts |
| West Hills       | 98.3%       | 94.5%                    | 3.8 pts |
| East Valley      | 97.9%       | 93.2%                    | 4.7 pts |

**East Valley's overall SLA rate (97.9%) looks unremarkable** — it would not trip an aggregate-level alert, and would even sit within a point of the field. The gap only becomes visible when the analysis is segmented to the tight-window (1-hour) urgent-response jobs, where East Valley trails every other region and posts the widest overall-to-tight SLA drop-off in the dataset. The underlying driver is geography: East Valley's average technician travel time is roughly triple Central District's, and this project's generator deliberately encoded that difference to make exactly this kind of aggregate-vs-segment discrepancy visible — a reminder that a single blended SLA number is not sufficient evidence of dispatch health.

## 06 · Finding 3 — Skill-driven rework

| SKILL LEVEL | FIRST-TIME-FIX | CALLBACK RATE | COMPLETED JOBS |
|-------------|----------------|---------------|----------------|
| Apprentice  | 78.4%          | 21.6%         | 6,477          |
| Journeyman  | 88.0%          | 12.0%         | 23,845         |
| Master      | 95.4%          | 4.6%          | 3,378          |

The gradient is clean and monotonic across all three skill tiers — Apprentices run a callback rate roughly 4.7x Master technicians' — and it is the single most consistent driver behind the "Low first-time-fix" and "High callback" flags in the priority queue below: 17 of the 25 technicians on December 2025's queue carry at least one of those two flags.

## 07 · The priority action queue as an operational control

Every KPI above rolls up into a single, ranked worklist — this is the artifact meant to actually change a dispatch manager's Monday, not just inform it. A technician appears here when they breach one or more thresholds in `Ref_KPITargets` ; the risk score is a simple count of breaches, and the recommended action is generated directly from which thresholds were crossed.

| TECHNICIAN     | REGION      | UTIL. | OVERTIME | FTF   | SLA    | CALLBACK | RISK | ACTION                                                                       |
|----------------|-------------|-------|----------|-------|--------|----------|------|------------------------------------------------------------------------------|
| Dakota Mendez  | South Metro | 50.6% | 7.4%     | 75.0% | 100.0% | 25.0%    | 3    | Underutilized on weekdays but carrying 7.4% overtime; low FTF; high callback |
| Emerson Larsen | South Metro | 56.8% | 6.4%     | 76.7% | 97.7%  | 23.3%    | 3    | Underutilized on weekdays but carrying 6.4% overtime; low FTF; high callback |
| Hayden Weber   | South Metro | 48.7% | 6.1%     | 85.7% | 97.6%  | 14.3%    | 2    | Underutilized on weekdays but carrying 6.1% overtime; high callback          |
| Robin Park     | West Hills  | 62.1% | 1.3%     | 79.5% | 95.5%  | 20.5%    | 2    | Low FTF; high callback                                                       |

+ 21 more, ranked by risk score — full list in the live dashboard and `excel/Ridgeline_KPI_Dashboard.xlsx`

Source: `dbo.vw_PriorityActionQueue` , reproduced independently by formula in the Excel `Priority_Action_Queue` sheet.

## 08 · Data quality: what was checked, and what it caught

Three anomaly types were deliberately injected into the synthetic dataset (~0.45–0.5% of completed jobs each) so the data-quality layer would have real, known-quantity issues to detect rather than a clean dataset with nothing to validate against:

| CHECK                         | RATE   | STATUS |
|-------------------------------|--------|--------|
| Timestamp inversion           | 0.452% | PASS   |
| Missing actuals               | 0.452% | PASS   |
| Extreme duration              | 0.452% | PASS   |
| Orphan dimension keys         | 0.000% | Clean  |
| Inconsistent completion flags | 0.000% | Clean  |

**Excel/SQL cross-check: exact match** on all three non-zero anomaly types (169/169 each) — two independently-written implementations of the same check, in two different tools, agreeing exactly.

**A real bug this cross-checking caught.** While validating the Priority Action Queue, the Excel formula layer and the SQL view disagreed on First-Time-Fix/SLA/Callback percentages for specific technicians — not by rounding, but by tens of percentage points. Root cause: `vw_TechnicianKPIMonthly` joined a technician's jobs without a date filter, so a technician's monthly quality metrics were silently averaged across their entire 2-year history rather than the selected month. Region-level KPIs were unaffected. The original UAT suite didn't catch it because its test fixture only ever had one month of data — nothing to leak across. The fix and a new regression test (UAT-06a/06b) are in `sql/04_kpi_views.sql` and `sql/06_uat_test_cases.sql` ; the full account is in `docs/data_validation_report.md` §5. This is the practical argument for building the same KPI logic twice, independently, rather than having Excel simply display SQL's output.

## 09 · Management recommendations

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DISPATCH MANAGER · THIS WEEK

### **Rebalance South Metro against North Metro before adding headcount anywhere**

South Metro's structural underutilization (averaging 47.2%, and never overlapping any other region's range across 24 months) should be addressed by demand-shifting or reallocating routing before authorizing any new hires in higher-intensity regions — the capacity gap is a distribution problem first, a headcount problem only if rebalancing doesn't close it.

REGIONAL OPS LEAD, EAST VALLEY · THIS MONTH

### **Review dispatch routing/travel radius specifically for 1-hour-SLA job types**

Because the gap doesn't show up in East Valley's overall SLA number, this requires deliberately segmenting the review to urgent-response jobs — a blended monthly SLA report would keep missing it.

TRAINING LEAD · NEXT QUARTER

### **Pair Apprentice technicians with Master technicians on a rotating basis**

The ~4.7x callback-rate gap between Apprentice and Master technicians is large and consistent enough to justify a structured pairing/mentorship program rather than one-off coaching — target the specific technicians flagged in the priority queue first.

ANALYTICS/BI OWNER · ONGOING

### **Keep the SQL and Excel KPI logic independent, and re-run the UAT suite after any view change**

Section 08's bug was caught specifically because two independent implementations existed to disagree with each other. Collapsing them into one (e.g., Excel importing SQL's pre-aggregated output) would remove that safety net.

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## 10 · Appendix — traceability, UAT, reproducibility

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**Requirements traceability:** 10 business requirements (BR-01–BR-10), each mapped to a KPI/field, its SQL/Excel implementation, and a specific validation test — full matrix in

`docs/requirements_traceability_matrix.md`, mirrored in the Excel workbook's `Traceability_UAT` sheet.

**UAT suite:** 8 test cases ( `sql/06_uat_test_cases.sql` ) run inside a transaction against a fully isolated fixture and roll back automatically — the production dataset is never modified by running it. All 8 currently PASS, including a regression test (UAT-06a/06b) added specifically to close the gap that let the Section 08 bug through.

**Reproducibility:** the entire pipeline — schema, synthetic data generator, data-quality views, KPI views, stored procedures, UAT suite, Excel workbook builder, and this dashboard's data exports — runs from the SQL scripts and the `excel/Build_Workbook.ps1` script in this repository. See `README.md` for exact commands.

**Live dashboard:** an interactive version of Sections 04–07 is published and linked from the project README.

Ridgeline Field Services is a fictional company. This document and all data within it are synthetic, produced for a portfolio analytics project. Prepared with SQL Server, Power BI, Microsoft Excel, and a custom HTML dashboard.